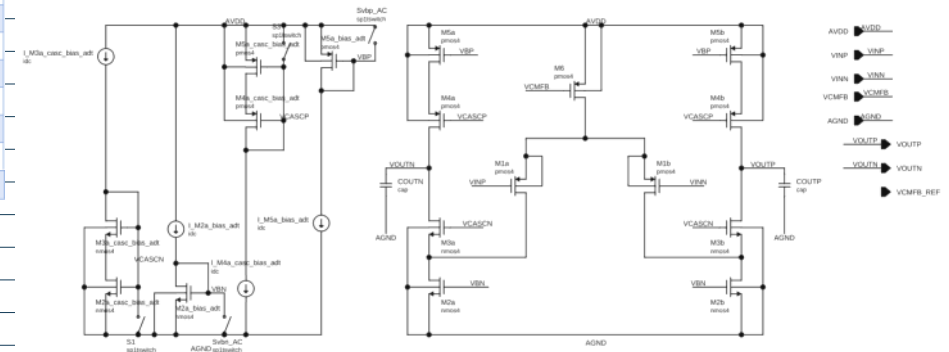


Technology	0.13um	GF180MCU
Supply voltage	1.2V	2.5V
Closed loop gain	2	2
Phase margin at the required ACL	$\geq 70^\circ$	$\geq 70^\circ$
CM input range – low	≤ 0	≤ 0
CM input range – high	$\geq 0.6V$	$\geq 1V$
Differential output swing	0.6Vpk-to-pk	1.2Vpk-to-pk
Load	500fF	500fF
DC Loop gain	50dB	60dB
CL settling time for 1% error	100ns	100ns



Circuit Analysis

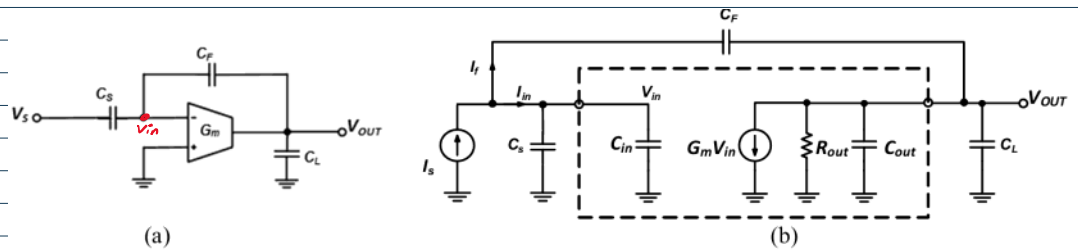


Fig. 2: (a) SC amplifier half-circuit in amplification phase and (b) Norton equivalent circuit.

1] G_m , R_{out} , C_{in} , C_L and Open Loop Gain— folded cascode :

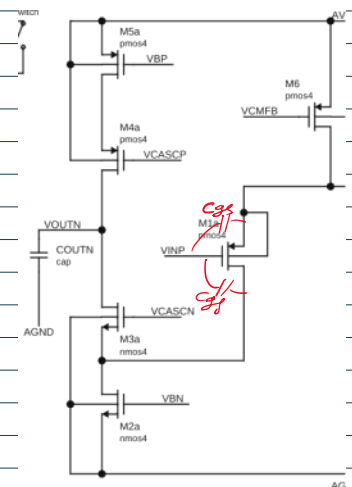
• $G_m = g_{m1/2} \times 8$ $\rightarrow$ less than 1 you need to put some margin to the $g_{m1/2}$

• $R_{out} = g_{m4} r_{o4} r_{o5} // g_{m3} r_{o3} \times [r_{o1} // r_{o2}]$

• $A_{v_{ol}} \geq 60 \text{ dB} = 1000$ $A_{v_{ol}} = G_m R_{out}$

$\rightarrow g_{m1} g_{m4} r_{o4} r_{o5} \geq 2000$ $\rightarrow g_{m1} g_{m3} r_{o3} (r_{o1} // r_{o2}) \geq 2000$

• $C_{in} = C_{gs1} + C_{gs3} (1 + g_{m1} \frac{1}{g_{m3}}) = C_{gs1} + 2 C_{gs3}$



2] Loop gain / closed-loop behavior :

$$\beta = \frac{V_{in}}{V_{act}} = \frac{C_F}{C_F + C_{in} + C_S} \quad \beta = \frac{V_{in}}{V_S} = \frac{C_S}{C_S + C_F + C_{in}}$$

$$\rightarrow |A_{cl}| = \frac{V_{out}}{V_S} = \frac{V_{out}}{V_{act}} \times \frac{V_{act}}{V_{in}} \times \frac{V_{in}}{V_S} = \frac{C_S}{C_F} = 2 \rightarrow C_S = 2 C_F$$

$$\rightarrow |A_{cl}| = \frac{V_{out}}{V_S} = \frac{V_{out}}{V_{in}} \times \frac{V_{in}}{V_S} = \frac{C_S}{C_F} = 2 \rightarrow C_S = 2C_F$$

$$\text{let } C_F \gg C_{in} \rightarrow \beta \approx \frac{C_F}{C_F + C_S} \approx \frac{1}{3}$$

$$\bullet B_{wd} = \frac{1}{\tau} = B \cdot GBW = B \cdot \frac{g_{m1}}{C_{out}}$$

$$\bullet V_{out} = V_{in}(1 - e^{-\frac{t}{\tau}}) \rightarrow \tau = \frac{t_{st}}{\ln(2)} = 21.7 \text{ ns}$$

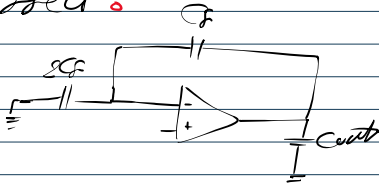
$t_{sp} = 4.6 \tau$

$$B_{wd} = 46 \text{ Mrad/sec} \rightarrow \frac{g_{m1}}{C} = 3B_{wd} = 138.18 \text{ Mrad/sec}$$

$\xrightarrow{\text{GBW}}$

$$C = C_{out} + \text{loading effect}$$

• loading effect :



$$\bullet C_L = C_{out} + \frac{2}{3}C_F \quad \text{let } C_F = 1 \text{ pF}$$

$$C_L = \frac{7}{6} \text{ pF}$$

$$\bullet \text{To Achieve settling Time} < 100 \text{ ns} \rightarrow g_{m1,2} \geq C_L \times 3B_{wd} = C_L \times \frac{13.8}{t_{st}} = 161 \mu\text{S}$$

3 Swing budget : $V_{out \text{ P-to-P}} = \text{Swing}_{\frac{1}{2}} = 0.6 \text{ V}$

$$\bullet V_{DS_{min}} = \frac{1}{4}(V_{DD} - 0.6) = 475 \text{ mV}$$

$$\bullet V_{out_CM} = 2V_{DS_{min}} + \frac{\text{swing}}{4} = 1.25 \text{ V}$$

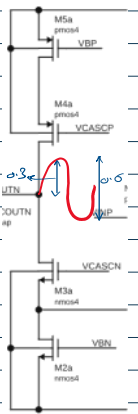
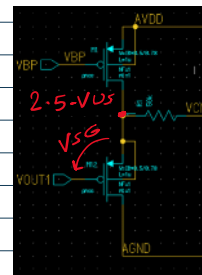
CMFB : If I Use NMOS CD

$$\bullet V_{out} \leq V_{DD} - V^* - V_{SG} \quad \bullet V_{out_max} = V_{out_cm} + 0.3 = 1.55 \rightarrow \text{too high}$$

Note: we need To Reduce V_{out_cm} abit to satisfy the V_{out_max} for the sensing circuit $\rightarrow$

$$\rightarrow \text{so let } \bullet V_{DS_{2,3}} = 425 \text{ mV} \quad \bullet V_{DS_{4,5}} = 525 \text{ mV}$$

$$\text{the new } V_{out_CM} = 2 \times V_{DS_{2,3}} + \frac{\text{swing}}{4} = 1.15 \text{ V}$$

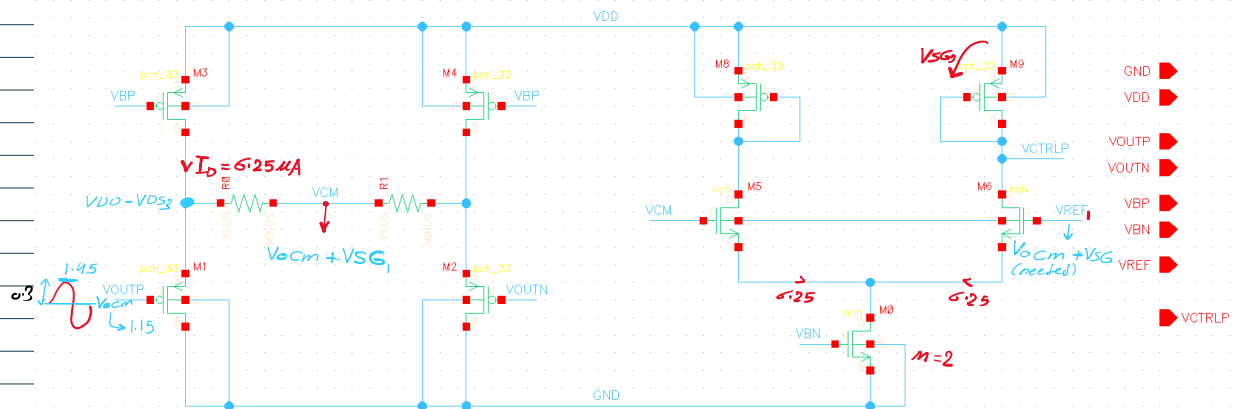


4 CMFB Analysis

$$V_{DD} - 0.5 \geq 1.45 + V_{SG1}$$

$$V_{DS_2} \leq 1.05 - V_{SG1}$$

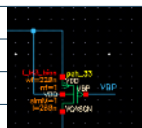
$$V_{SG1} \leq 1.05 - V_{SG1}$$



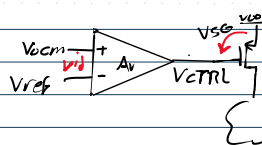
g_{m1} is ... V_{cm} ... V_{SG1} ... Note: ...

g_m is same as I_D
 $\rightarrow 10 \rightarrow V^* = 200 \text{ mV}$

$$V_{SG1} \leq 850 \text{ mV}$$



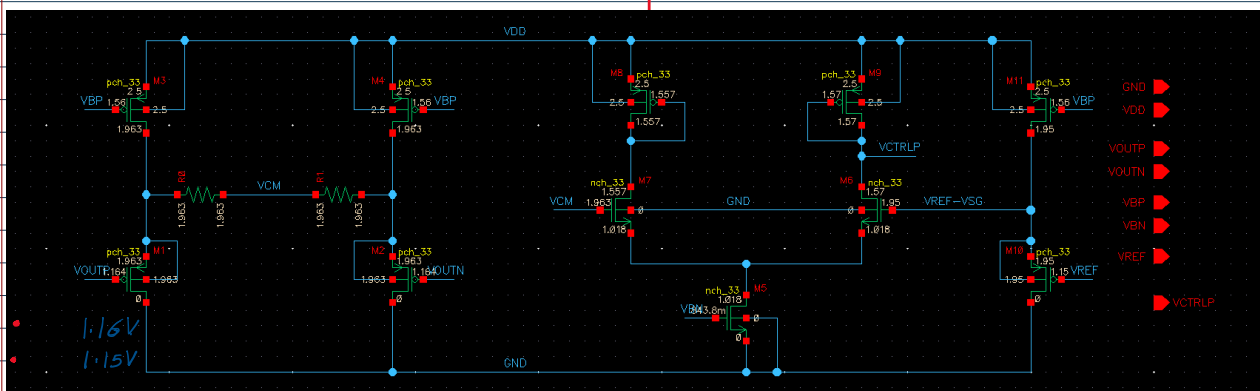
ID	6.25u
VGS	800m
I	1min
VDS	1.8
VSB	0
Stack	1
Results	[M1,2]
Name	TT-27.0
ID	6.25u
IG	N/A
L	280n
W	5.36u



Note: to Avoid having 2 HIN we designed the CMFB Amp to have LIN $\rightarrow$ low Gain ($A_v \approx 1$)

$V_{CTRL} = V_{OQ} + A_v V_{id}$ $V_{id} = V_{CTRL} - V_{OQ}$ we need small V_{id}

so we have to make $V_{OQ} = V_{CTRL} = V_{DD} - V_{SG}$ $V_{SG,10} = V_{SG}$ Tail



ID	6.25u
VGS	839.7m
lambda	100m
VDS	0.5
VSB	0
Stack	1
Results	TT-27.0
Name	TT-27.0
ID	6.25u
IG	N/A
L	450n
W	4.64u

DESIGN PROCEDURE

1 Input Pair [M1,2]

Specs $\rightarrow g_{m1} \geq 161 \mu\text{S}$

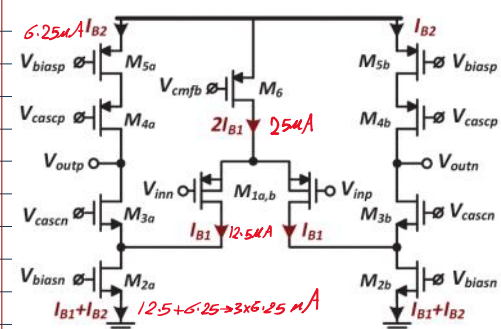
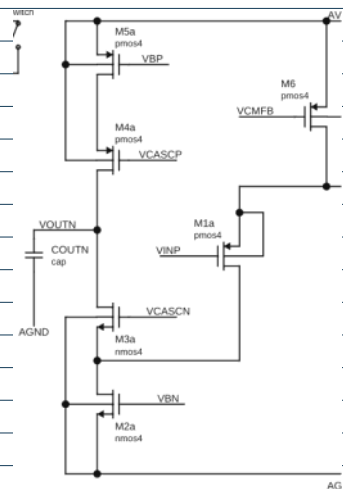
Gain $\rightarrow g_{m1} \cdot g_{m3} \cdot r_{o3} \cdot r_{o1} / 11 r_{o1} = 2 A_v$ let $r_{o1} = r_{o2}$ & $g_{m1} r_{o1} = g_{m3} r_{o3}$

$g_{m3} r_{o3} = g_{m1} r_{o1} = \sqrt{2 A_v} \rightarrow g_{m1} r_{o1} \geq 90$ $g_{m3} r_{o3} \approx 4.5$

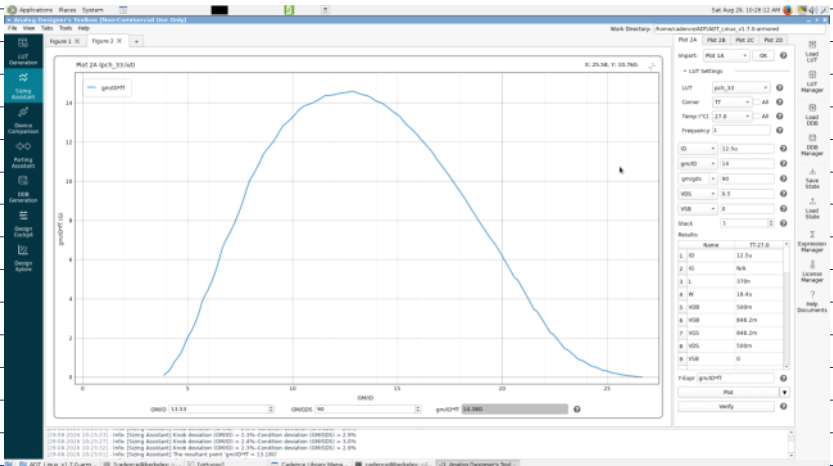
Note $\rightarrow$ we will use splitting Ratio (S) = 2 to get high GBW low noise & power

Disc. $\rightarrow$ Picked $g_m = 14$ To maximize the eff

& $I_D = 12.5$ To Achieve the min g_m



ID	12.5u
IG	N/A
L	370n
W	18.4u
VGS	848.2m
VDS	500m
VSB	0
VGB	848.2m
VDB	500m
gm/ID	13.92
Vstar	143.7m
ft	1.025G
gm/gds	84.68
VA	6.083
ID/W	679.3m
gm/W	9.457
AREA	6.808p
gm	174u
gmb	69.01u
gds	2.055u
ro	486.7k
Ron	40k
VTH	770.8m
VDSAT	121.7m
GB	27.01f



2] Cascode both N&B [M_3 & M_4] : $I_{D_{3,4}} = 6.25 \mu A$

$g_{m_{3,4}} r_{o_{3,4}} = \sqrt{2A} = 45 + \text{margin} = 50$

$V_{DS_{3min}} = 400m$ & $V_{DS_2} = 450$

$V_{DS_{n5,4}} = 525mV$

• we Picked $g_{m_{f_0}} = 10$ to Reduce Area

M_1

Import: Plot 1B

LUT Settings: pch_33

Corner: TT

Temp (°C): 27.0

Frequency: 1

ID: 6.25u

gm/ID: 10

gm/gds: 50

VDS: 0.4

VSB: 0.5

Stack: 1

Name	TT-27.0
1 ID	6.25u
2 IG	N/A
3 L	360n
4 W	3.6u

M_3

Import: Plot 1A

LUT Settings: nch_33

Corner: TT

Temp (°C): 27.0

Frequency: 1

ID: 6.25u

gm/ID: 10

gm/gds: 50

VDS: 0.4

VSB: 0.45

Stack: 1

Name	TT-27.0
1 ID	6.25u
2 IG	N/A
3 L	410n
4 W	1.28u

M_4

ID: 6.25u

IG: N/A

L: 360n

W: 3.6u

VDB: 900m

VGB: 1.596

VGS: 1.096

VDS: 400m

VSB: 500m

VGB: 1.596

VDB: 900m

gm/ID: 9.717

Vstar: 205.8m

ft: 1.857G

gm/gds: 49.25

VA: 5.069

ID/W: 1.736

gm/W: 16.87

AREA: 1.296p

gm: 60.73u

gmb: 18.3u

gds: 1.233u

ro: 811k

Ron: 64k

VTH: 939.6m

VDSAT: 190.2m

M_5

ID: 6.25u

IG: N/A

L: 450n

W: 1.28u

VDB: 900m

VGB: 1.596

VGS: 1.096

VDS: 400m

VSB: 500m

VGB: 1.596

VDB: 900m

gm/ID: 9.717

Vstar: 205.8m

ft: 1.857G

gm/gds: 49.81

VA: 5.043

ID/W: 4.883

gm/W: 48.22

AREA: 524.8f

gm: 61.73u

gmb: 15.58u

gds: 1.239u

ro: 806.9k

Ron: 64k

VTH: 822.8m

VDSAT: 171.6m

cgg: 1.855f

LUT Settings: pch_33

Corner: TT

Temp (°C): 27.0

Frequency: 1

ID: 25u

gm/ID: 10

lambda: 100m

VDS: 0.5

VSB: 0

Stack: 1

Name	TT-27.0
1 ID	25u
2 IG	N/A
3 L	450n
4 W	18.72u

ID: 25u

IG: N/A

L: 450n

W: 18.72u

VDB: 500m

VGB: 939.7m

VGS: 939.7m

VDS: 500m

VSB: 0

VGB: 939.7m

VDB: 500m

gm/ID: 9.906

Vstar: 201.9m

ft: 1.209G

gm/gds: 97.94

VA: 9.887

ID/W: 1.335

gm/W: 13.23

AREA: 8.424p

gm: 247.6u

gmb: 107u

gds: 2.529u

ro: 395.5k

Ron: 20k

VTH: 782.9m

VDSAT: 175m

same as I_{M_2} bias but with mult=4

3] Current sources [M_2 & M_5] :

M_2

$g_{m_1} r_{o_2} = 50$

$I_D = 12.5 + 6.25 = 3 \times 6.25$

$W_4 = M \times W_{p2-bias} = 4.8 \mu$

$M_2\text{-bias} \rightarrow W = 1.54 \mu \text{ \& } L = 550n$

LUT Settings: nch_33

Corner: TT

Temp (°C): 27.0

Frequency: 1

ID: 3*6.25u

gm/ID: 10

ro: 90/174u

VDS: 0.45

VSB: 0

Stack: 1

Name	TT-27.0
1 ID	18.75u
2 IG	N/A
3 L	550n
4 W	4.84u

M_5

$g_{m_1} r_{o_5} = \sqrt{2A} = 45 + \text{margin} = 50$

$I_{D_5} = 6.25 \mu A$

LUT Settings: pch_33

Corner: TT

Temp (°C): 27.0

Frequency: 1

ID: 6.25u

gm/ID: 10

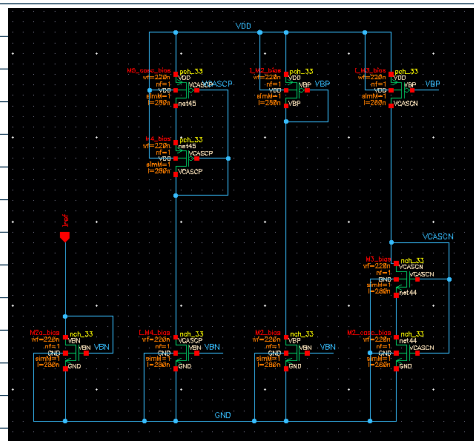
lambda: 100m

VDS: 0.5

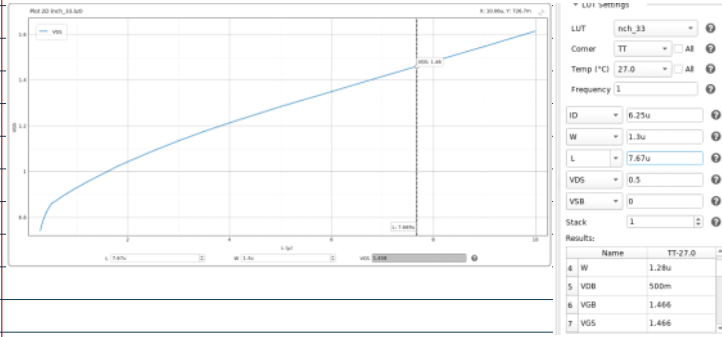
VSB: 0

Stack: 1

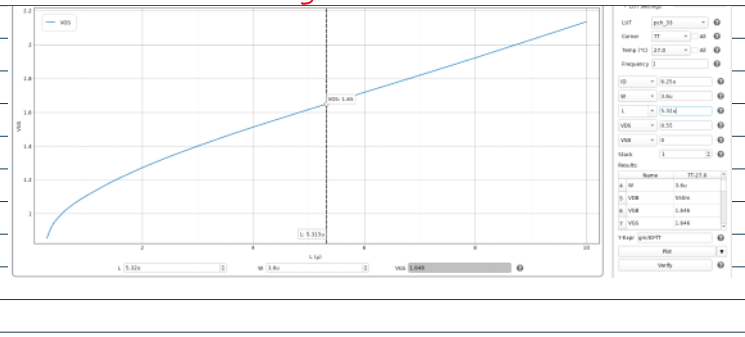
Name	TT-27.0
1 ID	6.25u
2 IG	N/A
3 L	450n
4 W	4.68u



M2-Casc-bias



M5-Casc-bias



M5

M5

M4

M3

M1

M2

ID 25u	ID 6.25u	ID 6.25u	ID 6.25u	ID 12.5u	ID 18.75u
IG N/A	IG N/A	IG N/A	IG N/A	IG N/A	IG N/A
L 450n	L 450n	L 360n	L 360n	L 370n	L 370n
W 18.72u	W 4.68u	W 3.6u	W 1.28u	W 18.4u	W 1.28u
VDB 500m	VDB 500m	VDB 900m	VDB 850m	VDB 500m	VDB 450m
VGB 939.7m	VGB 939.7m	VGB 1.596	VGB 1.408	VGB 848.2m	VGB 845.3m
VGS 939.7m	VGS 939.7m	VGS 1.096	VGS 957.7m	VGS 848.2m	VGS 845.3m
VDS 500m	VDS 500m	VDS 400m	VDS 400m	VDS 500m	VDS 450m
VSB 0	VSB 0	VSB 500m	VSB 450m	VSB 0	VSB 0
VGB 939.7m	VGB 939.7m	VGB 1.596	VGB 1.408	VGB 848.2m	VGB 845.3m
VDB 500m	VDB 500m	VDB 900m	VDB 850m	VDB 500m	VDB 450m
gm/ID 9.906	gm/ID 9.906	gm/ID 9.717	gm/ID 9.876	gm/ID 13.92	gm/ID 9.929
Vstar 201.9m	Vstar 201.9m	Vstar 205.8m	Vstar 202.5m	Vstar 143.7m	Vstar 201.4m
ft 1.209G	ft 1.209G	ft 1.857G	ft 5.295G	ft 1.025G	ft 3.382G
gm/gds 97.94	gm/gds 97.94	gm/gds 49.25	gm/gds 49.81	gm/gds 84.68	gm/gds 92.85
VA 9.887	VA 9.887	VA 5.069	VA 5.043	VA 6.083	VA 9.351
ID/W 1.335	ID/W 1.335	ID/W 1.736	ID/W 4.883	ID/W 679.3m	ID/W 3.874
gm/W 13.23	gm/W 13.23	gm/W 16.87	gm/W 48.22	gm/W 9.457	gm/W 38.46
AREA 8.424p	AREA 2.106p	AREA 1.296p	AREA 524.8f	AREA 6.808p	AREA 2.662p
gm 247.6u	gm 61.91u	gm 60.73u	gm 61.73u	gm 174u	gm 146.2u
gmb 107u	gmb 26.74u	gmb 18.3u	gmb 15.58u	gmb 69.01u	gmb 67.43u
gds 2.529u	gds 632.1n	gds 1.233u	gds 1.239u	gds 2.055u	gds 2.005u
ro 20k	ro 1.582MEG	ro 811k	ro 806.9k	ro 486.7k	ro 498.7k
VTH 782.9m	VTH 782.9m	VTH 939.6m	VTH 822.8m	VTH 770.8m	VTH 706.6m
VDSAT 175m	VDSAT 175m	VDSAT 190.2m	VDSAT 171.8m	VDSAT 121.7m	VDSAT 172.1m
			cgg 1.855f	cgg 27.01f	

